Advanced Topics in Computer Graphics II

Universität Bonn
Institut für Informatik II
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Sheet G11 - Editing and Boundary Constraints

Solutions for the theoretical and practical part via eCampus by Tue, 07.01.2025, **12:00**.

Practical Part

Assignment 1) Shell-based Editing *(4Pts)*

Download `exercise11Editing.zip` from eCampus and add it to the project. In the function `editMesh(...)`, construct the linear system of equations for solving the stretching and bending energy formulation from the lecture. Solve the system to obtain the displacement field of the vertices.

Assignment 2) Generalized Barycentric Mapping *(10Pts)*

Download `exercise11Parameterization.zip` from eCampus and add it to the project. In this task you will implement the barycentric mapping for parameterization.

a) In `detect_boundary_edges(...)`, detect boundary edges and return them.
b) In `detect_boundary_path(...)`, find the boundary path of consecutive vertices.
c) In `path_length(...)`, calculate the length of each boundary edge and store them inside a vector.
d) In `map_boundary_edges_to_circle(...)`, map the boundary edges onto a circle, such that the length ratio between the edges stays the same.
e) In `method_switcher(...)`, compute the weights for the different weighting methods.
Hint: Remember that `Eigen::SparseMatrix.setFromTriplets(...)` adds entries with the same indices.
f) In `construct_operator(...)`, build the operator for the linear system of equations. In the lecture, the system of equations was given as

$$\begin{pmatrix} 1 & -\lambda_{12} & \cdots & -\lambda_{1n} \\ \vdots & \ddots & \ddots & \vdots \\ -\lambda_{n1} & \cdots & -\lambda_{n(n-1)} & 1 \end{pmatrix} \begin{pmatrix} u_1 & v_1 \\ \vdots & \vdots \\ u_n & u_n \end{pmatrix} = \begin{pmatrix} \lambda_{1(n+1)} & \cdots & \lambda_{1N} \\ \vdots & \ddots & \vdots \\ \lambda_{n(n+1)} & \cdots & \lambda_{nN} \end{pmatrix} \begin{pmatrix} u_{n+1} & v_{n+1} \\ \vdots & \vdots \\ u_N & v_N \end{pmatrix}$$

where $i \in \{1, \ldots, n\}$ are the interior vertices and $i \in \{n+1, \ldots, N\}$ are the (known) boundary vertices. To solve this in practice, construct the linear system of equations

$$\begin{pmatrix} 1 & -\lambda_{12} & \cdots & -\lambda_{1n} & -\lambda_{1(n+1)} & \cdots & -\lambda_{1N} \\ \vdots & \ddots & \ddots & \vdots & \vdots & \ddots & \vdots \\ -\lambda_{n1} & \cdots & -\lambda_{n(n-1)} & 1 & -\lambda_{n(n+1)} & \cdots & -\lambda_{nN} \\ 0 & \cdots & \cdots & 0 & 1 & \cdots & 0 \\ \vdots & \ddots & \ddots & \vdots & \vdots & \ddots & \vdots \\ 0 & \cdots & \cdots & 0 & 0 & \cdots & 1 \end{pmatrix} \begin{pmatrix} u_1 & v_1 \\ \vdots & \vdots \\ u_n & u_n \\ u_{n+1} & v_{n+1} \\ \vdots & \vdots \\ u_N & u_N \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ \vdots & \vdots \\ 0 & 0 \\ u_{n+1} & v_{n+1} \\ \vdots & \vdots \\ u_N & v_N \end{pmatrix}$$

Note that in practice, we don't sort boundary vertices but we slice the corresponding rows of the matrix.

Theoretical Part

Assignment 3) Conformal Mapping *(4Pts)*

Explain the concept of conformal mappings. State and explain the three different classifications based on the singular values.

Good luck!